

Choosing a Model from the Evidence

linear models, quadratic models, and piecewise models

- Before you compute, decide which family the situation belongs to. Constant *first* differences point to a linear rule; constant *second* differences point to a quadratic rule; a rule that changes at a break point is piecewise.
 - Show the differences, the substitution, or the equation you solved. Write the final answer on the answer line.
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1. A shop prints team shirts. The table shows the total cost of an order.

Shirts ordered n	0	1	2	3
Total cost (dollars)	25	32	39	46

- (a) Is the total cost linear, quadratic, or piecewise? Explain which pattern in the table tells you.

Answer: _____

- (b) Find the total cost of an order of 10 shirts, in dollars.

Answer: _____

2. A stunt ramp launches a cart. Its height above the ramp, in feet, is recorded each second.

Time t (seconds)	0	1	2	3
Height (feet)	0	44	56	36

(a) Is the height linear, quadratic, or piecewise? Explain which pattern in the table tells you.

Answer: _____

(b) The height fits the rule $h(t) = -16t^2 + 60t$. Find the height at $t = 2.5$ seconds.

Answer: _____ ft

3. A garage charges \$3 for the first hour of parking and \$2 for every hour after that.

(a) Is the charge linear, quadratic, or piecewise? Explain what in the pricing tells you.

Answer: _____

(b) Find the charge for parking 4 hours, in dollars.

Answer: _____

4. A phone plan costs \$12 per month plus \$0.05 for each minute of talk time. One month the bill was \$27.50. Write and solve an equation for m , the number of minutes used.

Answer: _____ min

5. A model rocket is fired straight up from the ground. Its height in feet after t seconds is $h(t) = -16t^2 + 64t$. Solve $h(t) = 0$ to find the two times at which the rocket is at ground level.

Answer: _____ s

6. A tank fills at 5 litres per minute for the first 8 minutes, then at 2 litres per minute. Find the volume in the tank 15 minutes after filling starts.

Answer: _____ L

7. Gym A charges a \$40 joining fee plus \$15 per month. Gym B charges a \$25 joining fee plus \$20 per month. Find the number of months m and the total cost y at which the two gyms cost the same. Give both values.

Answer: _____

8. A rectangular dog run is fenced with 40 feet of fencing. If the width is w feet, the length is $20 - w$ feet, so the area is $A(w) = w(20 - w)$.

(a) Is the area linear, quadratic, or piecewise as a function of w ? Justify your answer from the rule.

Answer: _____

(b) Find the area when the width is 6 feet.

Answer: _____ ft^2

- 9.** A courier charges \$6 for a parcel of up to 5 pounds, then \$1.20 for each pound above 5. A parcel cost \$12 to send. Write and solve an equation for its weight w .

Answer: _____ lb

- 10.** A student looks at the table below and says “the rule must be linear, because y goes up every time.”

x	1	2	3	4
y	3	7	13	21

- (a) Explain why that reason is not enough, and name the family the table really belongs to.

Answer: _____

- (b) The table fits $y = x^2 + x + 1$. Predict y when $x = 5$.

Answer: _____

11. An electricity supplier charges \$0.11 per kilowatt-hour for the first 400 kilowatt-hours used in a month, and \$0.16 per kilowatt-hour after that. Find the bill, in dollars, for a month in which $k = 650$ kilowatt-hours were used.

Answer: _____

12. A farm sells crates of apples. When x hundred crates are sold, revenue is $R(x) = x(40 - 2x)$ hundred dollars and cost is $C(x) = 12x + 48$ hundred dollars.

(a) Name the family of each rule and justify each choice from the way the rule is built.

Answer: _____

(b) Solve $R(x) = C(x)$ to find both break-even values of x .

Answer: _____